Artificial Intelligence
Lecture 5: Multi-agent decision making

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Outline

1 Zero-sum games

2 Finite games

3 Opening thoughts: Player modeling

Plan

1. Zero-sum games
   - Motivation: When search is not enough
   - Matrix games
   - Strategic dominance
   - Defining strategies
   - Minimax and maximin strategies
2. Finite games
3. Opening thoughts: Player modeling

Last week...

We have seen that games such as chess, checkers, tic-tac-toe, are called **zero-sum games**, because *the winner wins what the loser loses*.

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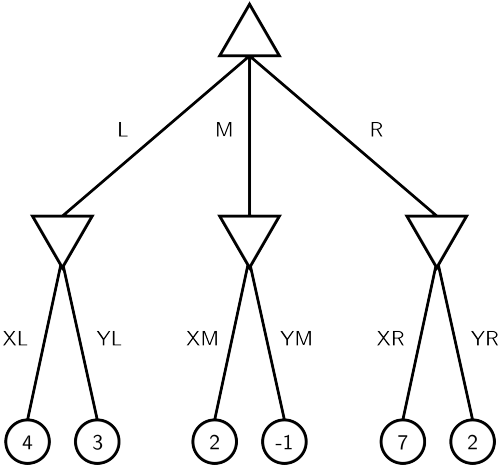
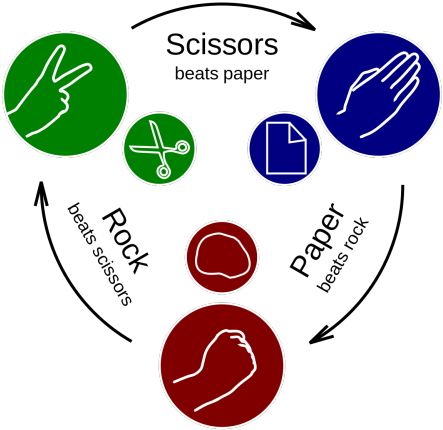
The algorithm used for finding the optimal strategy is called **Minimax**, and can be speed up using **alpha-beta pruning**.

An example of a game tree



A simple game: Rock-Paper-Scissor



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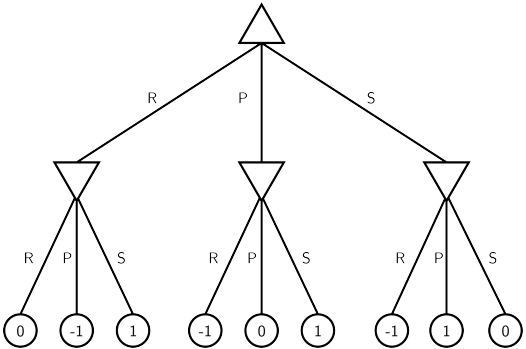
Question: Can you represent this game in the form of a tree?



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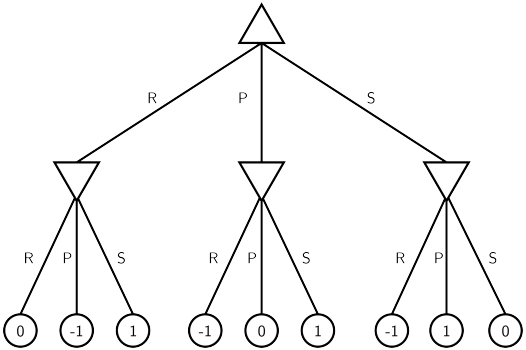
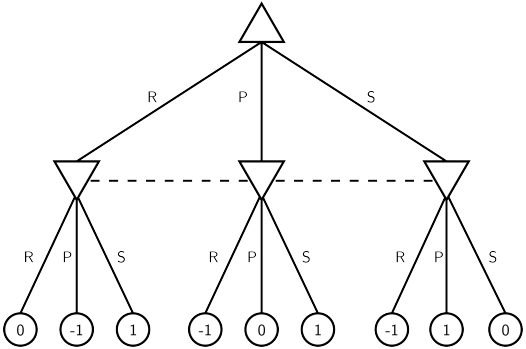


No! It is not possible here (at least not directly), since the actions are **simultaneous**. Player 2 does not wait until player 1 has played to play.

A solution for simultaneous actions: No observability

A solution to this is to consider that Player 2 has **no observability** over the action of Player 1.



Rock-Paper-Scissor: Optimal action?

When Player 2 can see the action of Player 1, Player 2 has an optimal strategy.

How does non-observability affect this result?

Rock-Paper-Scissor: Optimal action?

When Player 2 can see the action of Player 1, Player 2 has an optimal strategy.

How does non-observability affect this result?

Now there is no optimal action anymore for Player 2: whatever P2 does, P1 may have chosen an action that makes P2 lose.

Conversely, P1 cannot make any action that would *for sure* win over any action of P2.

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Matrix representation of the Rock-Paper-Scissor game

Since the actions of both players are simultaneous, we can represent the outcome of the Rock-Paper-Scissor game in the form of a matrix:

		Player 2		
		R	P	S
Player 1	R	0	-1	1
	P	1	0	-1
	S	-1	1	0

Matrix representation of zero-sum games

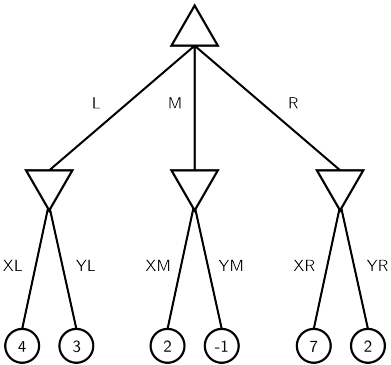
A common way of representing zero-sum games is to use a matrix G such that:

- The rows of G are the actions of Player 1
- The columns of G are the actions of Player 2
- g_{ij} is the payoff of Player 1 when P1 plays action i and P2 plays action j.

Remark: The payoff of Player 2 is then $-g_{ij}$

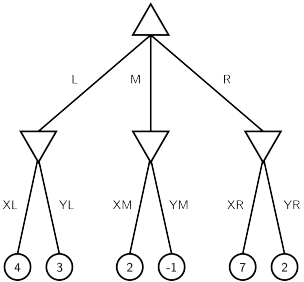
Back to game trees?

The matrix form works naturally in a context where players play one single action simultaneously. Can it describe a sequential interaction?



Matrix representation of a game tree

Player 1 \ Player 2	XL, XM, XR	XL, XM, YR	XL, YM, XR	XL, YM, YR	YL, XM, XR	YL, XM, YR	YL, YM, XR	YL, YM, YR
L	4	4	4	4	3	3	3	3
M	2	2	-1	-1	2	2	-1	-1
R	7	2	7	2	7	2	7	2



Strategies for Player 1: (L, M, R)

Strategies for Player 2: The strategies are conditional on the branch in the tree.

For instance, the strategy (XL, YM, XR) of Player 2 reads:

- If Player 1 chooses L, play XL
- If Player 1 chooses M, play YM
- If Player 1 chooses R, play XR

Normal-form vs Extensive-form

The **normal form** of the game is the matrix representation of the game, where g_{ij} is the payoff of Player 1 when Player 1 plays i and Player 2 plays j.

The **extensive form** of the game is a graphical representation of the game, where actions are represented by some game tree, incorporating the representation of sequential actions, partial observability and chance.

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Another example

Consider the following game:

		Player 2		
		a	b	c
Player 1	A	1	-1	1
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What do you observe about strategy B?

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What do you observe about strategy B?

Whatever Player 2 may play, the payoff of Player 1 will always be higher if playing B. We say that B is a **dominant** strategy for Player 1.

Another example

Consider the following game:

		Player 2		
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Does Player 2 have any dominant strategy?

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Consider the following game:

		Player 2		
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Player 1	A	1	-1	1
	B	2	1	1
	C	1	0	0

Does Player 2 have any dominant strategy?

Whatever Player 1 may play, the payoff of Player 2 will always be higher if playing b. Therefore b is a **dominant** strategy for Player 2.

Dominant strategy for Rock-Paper-Scissor

		Player 2		
		R	P	S
Player 1	R	0	-1	1
	P	1	0	-1
	S	-1	1	0

Does Player 1 have any dominant strategy at the Rock-Paper-Scissor game?

Dominant strategy for Rock-Paper-Scissor

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Does Player 1 have any dominant strategy at the Rock-Paper-Scissor game?

No, at least not with this definition of strategies.

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Pure strategies

Until now, we have considered that a strategy corresponds to the choice of an action (potentially conditional to the state). Such a strategy is called a **pure** strategy.

In normal form games:

- The pure strategies of Player 1 correspond to the rows of the matrix
- The pure strategies of Player 2 correspond to the columns of the matrix

Mixed strategies

Which strategy would you play for Rock-Paper-Scissor?

Mixed strategies

Which strategy would you play for Rock-Paper-Scissor?

A strategy which seems reasonable is to play actions randomly. Such a strategy is called a **mixed strategy**.

Mixed strategies

Given a matrix game $G \in \mathbb{R}^{m \times n}$:

A **mixed strategy** for Player 1 is a probability distribution over rows of G, i.e. a vector $p \in \mathbb{R}^m$ such that:

$$\sum_{i=1}^{m} p_i = 1$$

A **mixed strategy** for Player 2 is a probability distribution over columns of G, i.e. a vector $q \in \mathbb{R}^n$ such that:

$$\sum_{j=1}^{n} q_j = 1$$

Remark: A pure strategy is just a specific case of mixed strategy, where $p_i = 1$ (resp. $q_j = 1$) for the specific action to play and $p_i = 0$ (resp. $q_j = 0$) otherwise.

Expected payoff

For pure strategies: If Player 1 plays i and Player 2 plays j, the payoff for Player 1 is g_{ij}.

Expected payoff

For pure strategies: If Player 1 plays i and Player 2 plays j, the payoff for Player 1 is g_{ij}.

For mixed strategies: If Player 1 has strategy p and Player 2 has strategy q:

- Player 1 plays i with probability p_i
- Player 2 plays j with probability q_j
- The expected value of the payoff for Player 1 is:

$$\begin{aligned}\mathbb{E}[V] &= \sum_{i=1}^{m}\sum_{j=1}^{n} p(\text{Player 1 plays } i)p(\text{Player 2 plays } j)g_{ij} \\ &= \sum_{i=1}^{m}\sum_{j=1}^{n} p_i g_{ij} q_j \\ &= p^T G q\end{aligned}$$

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Minimax strategy

We can generalize the notion of minimax and maximin (used for extensive form games) to the case of normal form games.

Reminder: Minimax algorithm computes the value of the optimal strategy for Player 1, under the assumption that Player 2 plays optimally.

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In a normal form game: The exact same definition applies!

If Player 1 has strategy p, it can guarantee an expected payoff of at least:

$$v_1(p) = \min_{q \in \Delta^n} p^T G q$$

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$$v_1(G) = \max_{p \in \Delta^m} v_1(p) = \max_{p \in \Delta^m} \min_{q \in \Delta^n} p^T G q$$

Minimax and maximin

In the best case, Player 1 can guarantee an expected payoff of at least:

$$v_1(G) = \max_{p \in \Delta^m} \min_{q \in \Delta^n} p^T G q$$

Similarly, Player 2 can guarantee to lose at most

$$v_2(G) = \min_{q \in \Delta^n} \max_{p \in \Delta^m} p^T G q$$

Minimax theorem

Theorem: Minimax Theorem for Matrix Games

For any matrix $G \in \mathbb{R}^{m \times n}$, we have:

$$v_1(G) = v_2(G)$$

i.e.

$$\max_{p \in \Delta^m} \min_{q \in \Delta^n} p^T G q = \min_{q \in \Delta^n} \max_{p \in \Delta^m} p^T G q$$

This quantity is called the **value** of the game.

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Bi-matrix games

In zero-sum games, the payoffs for Player 1 and Player 2 are strongly correlated:

$$U_1 + U_2 = 0$$

If knowing one, one knows the other. Consequently, we could represent the normal form of the game with **one** matrix.

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When the two utility functions are unconstrained, it is necessary to declare the payoffs for each agent, i.e. to define **two** matrices.

Bi-matrix games

A **bi-matrix game** is given by two matrices (A, B) of same dimensions: $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{m \times n}$:

- The rows of A and B correspond to the pure strategies of Player 1.
- The columns of A and B correspond to the pure strategies of Player 2.
- a_{ij} is the payoff of Player 1 if Player plays i and Player 2 plays j.
- b_{ij} is the payoff of Player 2 if Player plays i and Player 2 plays j.

Bi-matrix games: Notation

In general, we represent bi-matrix games with one single table, each cell containing two values.

		Player 2		
		a	b	c
Player 1	A	1, 3	-1, 1	1, -1
	B	2, 2	1, 5	1, 0
	C	1, -1	0, 5	0, -2

Dominating strategies

		Player 2		
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What can be said about Strategy B of Player 1?

Dominating strategies

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What can be said about Strategy B of Player 1?
Whatever Player 2 does, B is always better than A and C. We say that strategy B is a **dominating** strategy.

Dominating strategies

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		a	b	c
Player 1	A	1, 3	-1, 1	1, -1
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What can be said about Strategy B of Player 1?
Whatever Player 2 does, B is always better than A and C. We say that strategy B is a **dominating** strategy.

What can be said about Strategy c of Player 2?

Dominating strategies

		Player 2		
		a	b	c
Player 1	A	1, 3	-1, 1	1, -1
	B	2, 2	1, 5	1, 0
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What can be said about Strategy c of Player 2?
Whatever Player 1 does, c is always worse than a and b. We say that strategy c is a **dominated** strategy.

Mixed strategies in bi-matrix games

Such as for matrix games, we can define **mixed strategies** in bi-matrix games, as a probability distribution over pure strategies of the player.

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Such as for matrix games, we can define **mixed strategies** in bi-matrix games, as a probability distribution over pure strategies of the player.

If Player 1 plays mixed strategy p and Player 2 plays mixed strategy q:

- The expected payoff of Player 1 is:

$$p^T A q = \sum_{i=1}^{m} \sum_{j=1}^{n} p_i q_j a_{ij}$$

- The expected payoff of Player 2 is:

$$p^T B q = \sum_{i=1}^{m} \sum_{j=1}^{n} p_i q_j b_{ij}$$

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Best reply

The key question in game theory is:

Which strategy should I play to maximize my payoff?

In particular, answering to this question requires to answer to another related question:

Knowing how the other player plays, what should I play to maximize my payoff?

Best reply

A strategy p of Player 1 is a **best reply** to a strategy q of Player 2 if

$$p^T Aq \geq p'^T Aq$$

for all mixed strategy p'.

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Remarks:

- A dominating strategy is a best reply to all possible strategies of Player 2.
- The definition can naturally be adapted for strategies of Player 2.

Limitations of the concept of best reply

Best replies are defined for a **specific** and **known** strategy of the other player.

The problem is that we don't know which strategy Player 2 will play!

Example: In the Rock-Paper-Scissor game, if Player 2 decides to play scissor, the best reply of Player 1 is to player rock.

If we don't know what Player 2 will play, it is *a priori* difficult to know what to play.

Analogy with minimax

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Here, Player 1 can make the same assumption and assume that **Player 2 will aim to find a best response to Player 1's strategy**.

This situation where each player's strategy is a best response to the other player's strategy is called a **Nash equilibrium**.

Computing Nash equilibria in pure strategy

Consider the following game:

$$\begin{pmatrix} 2,2 & 4,0 & 1,1 & 3,2 \\ 0,3 & 1,5 & 4,4 & 3,4 \\ 2,0 & 2,1 & 5,1 & 1,0 \end{pmatrix}$$

What are the Nash equilibria in pure strategy for this game?

Computing Nash equilibria in mixed strategy

Consider the following game:

$$\begin{pmatrix} 2,2 & 0,1 & 1,0 \\ 1,0 & -2,1 & 0,-1 \\ 1,1 & 3,3 & 4,-5 \end{pmatrix}$$

What are the Nash equilibria in mixed strategy for this game?

The prisoner's dilemma

Two prisoners are being interrogated and have the choice between **cooperating** with their partner or **betraying**. They can't observe how the other is behaving.

Based on what their behaviour, they are sentenced to serve a different number of years in prison:

		Player 2	
		Cooperates	Defects
Player 1	Cooperates	(-1, -1)	(-3, 0)
	Defects	(0, -3)	(-2, -2)

The prisoner's dilemma

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The strategy (Defects, Defects) is the only Nash equilibrium in pure strategy.

However the strategy (Cooperates, Cooperates) leads to higher payoff for both agents. We say that (Defects, Defects) is not **Pareto optimal**: it does not lead to a maximal payoff for all agents.

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General definition

A **game in normal form** is defined by:

- A set of players: $\mathcal{P} = \{1, \ldots, r\}$
- For each player p:
  - A finite set S_p of pure strategies
  - A utility/payoff function $u_p : S_1 \times \ldots \times S_r \to \mathbb{R}$

A **mixed strategy** for player p is a probability distribution over S_p.

The **expected payoff** of Player p for the mixed strategy profile $\pi = (\pi_1, \ldots, \pi_r)$ is given by:

$$U_p(\pi_1, \ldots, \pi_r) = \sum_{s_1, \ldots, s_r} \pi_1^{s_1} \ldots \pi_r^{s_r} u_p(s_1, \ldots, s_r)$$

Existence of Nash equilibria in pure strategy

There does **not** always exist Nash equilibria in pure strategy.

This can be observed for instance in the following 2-player game:

$$\begin{pmatrix} 1,2 & 3,1 \\ 3,1 & 2,2 \end{pmatrix}$$

Existence of Nash equilibria for finite games

An important theoretical result of Game Theory is the existence of Nash equilibria for finite games:

Theorem

Every game with a **finite number of players** and a **finite number of actions** for each player has at least one Nash equilibrium in mixed strategy.

This is not (always) the case with an infinite number of players or when the number of actions is infinite.

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The main problem with game theory

The solutions to a game strongly depend on the strategy of the other players: we want to play a **best response** to their strategy.

In practice, this requires to answer two questions:

- What is the utility function of the other players?
- What is the other players' rationality?

Defining the utility: not so simple?

A salesperson wants to sell cookies in the street. They need to decide of a price for the cookies (from 0 euros to 10 euros per cookie).

Pedestrians decide how many cookies to buy, based on the price.

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Pedestrians decide how many cookies to buy, based on the price.

Rational choice for the pedestrians: If the cookies are free, take them all!

Irrational behaviours: Allais paradox

You are given the choice between the four lotteries:

- A: Win 4000 euros with probability 0.8
- B: Win 3000 euros for sure
- C: Win 4000 euros with probability 0.2
- D: Win 3000 euros with probability 0.25

Most people prefer B over A, and C over D, which is not consistent with the utilities

To know more...

Game theory:

- Norvig and Russell, chapter 18 (4th edition)
- Bonanno, G. (2024). Game theory

The roots of game theory:

- Kjeldsen, T. H. (2001). John von Neumann's conception of the minimax theorem: A journey through different mathematical contexts.
- Nash, J. F. (1951). Non-cooperative games.

Prisoner's dilemma:

- Raihani, N. J., & Bshary, R. (2011). Resolving the iterated prisoner's dilemma: theory and reality.

Human biases:

- Kahneman, D., & Tversky, A. (2013). Prospect theory: An analysis of decision under risk.
- Kahneman, D. (2011). Thinking, fast and slow.